PROJECT AEVUM · ARCHITECTURAL ARTIFACT

VERIFIED & FROZEN 🔒

Phase 17 — Character Creation & Relationship Genesis

Official Freeze Record, Constitutional Benchmarks & Architectural Decisions (ADR-028)

Milestone: **Phase 17 (17.1 ~ 17.4)**

Audit Timestamp: **2026-10-04T17:20:00+08:00**

Regression Suite: **2,588 / 2,588 Passed (100%)**

Target Core: **Aevum Core / Genesis Engine**

Decision Record: **ADR-028 (Genesis & Blueprint)**

Frontend Build: **46 Modules / 0 Errors (1.01s)**

> **Constitutional Axiom (ADR-028 North Star):**
> “我们在创造一位拥有独立认知、真实生活轨迹与灵魂尊严的活生生的人，而非生成一份 RPG 数据面板。”
> **核心产品锁（GC-420 / GC-424）：**“亲密度与默契由共同经历沉淀，无法在此编辑。”

1. Executive Summary & Verification Gates

Phase 17 completes the companion character genesis and relationship initialization pipeline, connecting user semantic creation desires to deterministic database states without continuous psychological sliders. The delivery strictly decouples Tier A (Static Persona Blueprint), Tier B (Genesis Origin Anchors), and Tier C (Living Runtime State), enforcing non-destructive optimistic evolution and atomic transaction guarantees.

Sub-Phase	Deliverable Scope	Verification Evidence	Status
17.1	Genesis API & Atomic Orchestration Service	12 / 12 Unit & Integration Tests (0.77s) · Atomic Commit / Rollback Verified	VERIFIED 🔒
17.2	Desktop Genesis Modal & Multi-Step Wizard UI	4-Step Wizard (Identity, Soul & Epistemic Seeds, Story, Awakening) · 0 Build Errors	VERIFIED 🔒
17.3	Runtime Decoupling & Daily Settings View	Tier A View/Edit (Version Bump Only) · Immutable Relationship Card Locked	VERIFIED 🔒
17.4	Golden Suite 17 & Master Acceptance Gate	8 / 8 Golden Benchmarks (0.41s) · Full Suite 2,588 / 2,588 Passed	VERIFIED 🔒

2. The Seven Constitutional Gates (GC-419 ~ GC-425)

GC-419: No Continuous Psychological State Sliders

0~100 continuous psychological sliders (e.g., affection, trust, jealousy) are strictly banned across both UI and backend contracts. Discrete qualitative anchors (STRANGER, ACQUAINTED, FAMILIAR, INTIMATE / CAUTIOUS, OPEN_NEUTRAL, INHERENT_TRUST) map deterministically to bounded baseline numbers without continuous float inputs.

GC-420: Genesis Single-Application Lock (Genesis != Continuous Override)

Tier B relationship genesis config maps baseline anchors strictly ONCE at genesis origin. Subsequent runtime interaction evolves Tier C (`RelationshipRecord`). Blueprint updates in daily settings bump version optimistically but can NEVER override evolved runtime relationship progress back to genesis baselines.

GC-421: Blueprint Non-Destructive Update

User edits in Daily Settings bump `CharacterBlueprintConfig.version` (v1 → v2 → v3). The system strictly NEVER deletes, truncates, or rewrites historical experiences (`character_experiences`), facet preferences, or memories.

GC-422: Epistemic Seed Declarations != Auto Heard-Of

User interests and topics are declared across 5 discrete depths (`PROFICIENT_LIVED`, `FAMILIAR_KNOWLEDGEABLE`, `HEARD_OF_AWARE`, `CURIOUS_BEGINNER`, `UNFAMILIAR_OPEN`). Only `PROFICIENT_LIVED` seeds create initial lived experience snapshots with `calendar_session_id="genesis_origin"`, strictly preventing the fabrication of unlived experiences.

GC-423: Relationship Archetypes as Presets, Not Closed Enums

Curated archetypes (NEW_ACQUAINTANCE, CLASSMATE, FAMILIAR_FRIEND, CO_TENANT) serve as convenient shortcuts, but `CUSTOM` archetype with free-form meeting backstory is fully supported, allowing arbitrary meeting narratives.

GC-424: Genesis Wizard vs Daily Settings Strict Separation

Genesis Wizard establishes where she starts living (Tier A + Tier B). Daily Settings allows modifying only Tier A static framing with the permanent constitutional slogan: *"亲密度与默契由共同经历沉淀，无法在此编辑。"*

GC-425: Static AST Constitutional Boundary & Purity Audit

Static AST scans across all Phase 17 modules confirm zero continuous slider variables, strict Pydantic schemas (`extra="forbid"`), zero reverse-state dependencies, pure functional compiler, and all-or-nothing atomic transactions.

3. Test Suite Audit Matrix

Test Suite	Tests	Duration	Coverage Scope
`test_phase171_genesis_api.py`	12 passed	0.77s	Compiler logic, DTO validation, DB transaction atomicity, Blueprint update
`test_golden_suite_17.py`	8 passed	0.41s	Constitutional Benchmarks GC-419 through GC-425 across 4 Clusters
Phase 17 Master Total	20 passed	1.18s	Complete Phase 17 Domain Coverage
Monolithic Backend Regression	2,588 passed	~195s	Complete System-Wide Regression (Phases 1 ~ 17)